Synthesis and Characterization of Organo-Soluble Polyimides Based on Polycondensation Chemistry of Fluorene-Containing Dianhydride and Amide-Bridged Diamines with Good Optical Transparency and Glass Transition Temperatures over 400 °C

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Abstract

A series of wholly aromatic polyimide (PI) films were prepared via polycondensation of one fluorene-containing dianhydride 9,9-bis(3,4-dicarboxyphenyl)fluorene dianhydride (FDAn) and several aromatic diamines with amide linkages. The resulting PI films showed Tg values of 422–436 °C, more than 35 °C higher than the 6FDA-PI counterparts. The optical cutoff wavelengths (λ■) were below 380 nm and transmittances above 80% at 450 nm.

2. Materials and Methods

All reactions were conducted under nitrogen atmosphere. N,N-dimethylacetamide (DMAc) was used as solvent. The poly(amic acid) (PAA) precursors were prepared by reacting equimolar amounts of dianhydride and diamine at room temperature for 12 h. Thermal imidization was performed at 100 °C (1 h), 200 °C (1 h), and 300 °C (1 h). Films were cast on glass plates.

3. Results and Discussion

The Tg values were measured by differential scanning calorimetry (DSC) under nitrogen atmosphere at a heating rate of 10 °C/min. The second heating cycle was used for Tg determination. Thermogravimetric analysis (TGA) was performed at 10 °C/min under nitrogen.

FLPI-1 (FDAn/FDAADA) exhibited Tg = 436.4 °C and Td5% = 524 °C in nitrogen. All FDAn-based polyimides showed Tg values in the range of 422–436 °C, significantly higher than the corresponding 6FDA-based polyimides (385–401 °C). This enhancement is attributed to the rigid fluorene unit restricting chain mobility.

Table 4. Thermal properties of polyimide films.

Sample ID	Dianhydride	Diamine	Tg (°C) (DSC)	Td5% N■ (°C)	Td5% air (°C)	Char yield 800°C (%)
FLPI-1	FDAn	FDAADA	436.4	524	498	62
FLPI-2	FDAn	ABTFMB	428.7	519	491	58
FLPI-3	FDAn	TFMB	422.2	510	485	55
6FDA-PI-1	6FDA	FDAADA	401.3	498	472	51
6FDA-PI-2	6FDA	ABTFMB	393.5	492	465	48
6FDA-PI-3	6FDA	TFMB	385.1	485	458	45

Note: Tg = glass transition temperature; Td5% = temperature of 5% weight loss; Char yield = residual weight at 800 °C in nitrogen. FDAn = 9,9-bis(3,4-dicarboxyphenyl)fluorene dianhydride; 6FDA = 4,4′-(hexafluoroisopropylidene)diphthalic anhydride; FDAADA = 9,9-bis[4-(4-aminobenzamide)phenyl]fluorene; ABTFMB = 2,2′-bis(trifluoromethyl)-4,4′-bis[4-(4-aminobenzamide)]biphenyl; TFMB = 2,2′-bis(trifluoromethyl)-4,4′-diaminobiphenyl.

3.2 Optical Properties

The optical cutoff wavelength (λ■) of all FDAn-PI films is below 380 nm. The transmittance at 450 nm (T■■■) exceeds 80% for all films. The yellow index (YI) values are all below 12, indicating good colorlessness. These optical properties are maintained even after aging at 200 °C for 100 h.